



The 3rd International Mathematical Summer Camp (IMSC2025)

Competition-Day II

Date: Sunday, July 6th, 2025

Time: 09:30-14:00

Number of Problems: 3

Total Points: 21

Problem 4. Find all positive integers n such that there exists a convex n -gon that can be cut into a finite number of (not necessarily congruent) triangles each with angles equal to 15° , 75° , and 90° .

Problem 5. Let $Q(x)$ be a nonzero polynomial with integer coefficients. A positive integer n is *special* if for every prime number p , the following holds:

$$p \mid \sigma(n) + 1 \quad \text{if and only if} \quad p \mid Q(n).$$

Show that there are infinitely many positive integers n such that neither n nor $n + 1$ is *special*.

Note: Here $\sigma(n)$ denotes the sum of the positive divisors of n .

Problem 6. Let ABC be a scalene acute-angled triangle with orthocentre H and circumcircle ω . Let M be the midpoint of the arc BC of ω that contains A . Let lines BM and CH meet at E , and lines CM and BH meet at F . Lines BC and EF meet at K . If L is the projection of M onto line KH , prove that $AH = AL$.